
Migrating within STM32H5 MCUs

Introduction

The designers of STM32 microcontroller applications must have the possibility to replace one microcontroller type with another one from the same product family or products from a different family easily. The reasons for migrating an application to a different microcontroller can be for example:

- To fulfill higher product requirements, extra demands on memory size, or an increased number of I/Os
- To meet the cost reduction constraints that require to switch to smaller components and shrink the PCB area.

This application note details the steps required to migrate existing design between STM32H5 series microcontrollers.

This document provides the full set of features available for the STM32H503, STM32H523/533, STM32H543/553, STM32H562, STM32H563/573, and STM32H5Ex/STM32H5Fx devices. This document also provides guidelines on both hardware and peripheral migration.

All the above devices are referred to as STM32H5 unless otherwise stated.

Migrating between devices within the same family could require hardware or software changes in some cases. Such changes are described in this document.

To understand better the information in this application note, the user must be familiar with the STM32 microcontroller family.

For additional information, refer to the product datasheets and reference manuals.

1 General information

STM32H503, STM32H523/533, STM32H543/553, STM32H562, STM32H563/573, and STM32H5Ex/STM32H5Fx MCUs are 32-bit microcontrollers based on the Arm® Cortex® processor.

arm

*Note: Arm and Cortex are registered trademarks of Arm Limited (or its subsidiaries or affiliates) in the US and/or elsewhere.
 The Arm word and logo are trademarks of Arm Limited (or its subsidiaries) in the US and/or elsewhere. All rights reserved.*

Table 1. Reference documents

Document number	Title
[1]	STM32H523/33xx, STM32H543/553, STM32H562/63xx, and STM32H573xx Arm®-based 32-bit MCUs reference manual (RM0481)
[2]	STM32H503 line Arm®-based 32-bit MCUs reference manual (RM0492)
[3]	STM32H5E4/5F4 and STM32H5E5/5F5 Arm®-based 32-bit MCUs reference manual (RM0517)
[4]	Getting started with STM32H5 MCU hardware development (AN5711)

2 STM32H5 MCU overview

This section covers the following devices:

- STM32H503
- STM32H523/533
- STM32H543/553
- STM32H562
- STM32H563/573
- STM32H5Ex/STM32H5Fx

2.1 Memory availability

Table 2 summarizes the memory availability of STM32H503, STM32H523/533, STM32H543/553, STM32H562, STM32H563/573, and STM32H5Ex/STM32H5Fx MCUs.

Table 2. Memory availability

Products	Flash memory		RAM size (Kbytes)						Feature level
	Size	Bank	SRAM1	SRAM2	SRAM3	SRAM4	SRAM5	BKPSRAM	
STM32H5Fx	4 Mbytes	Dual	256	128	384	384	384	4	With hardware crypto: AES, PKA, SAES, and OTFDEC
STM32H5Ex	Up to 4 Mbytes	Dual	256	128	384	384	384	4	N/A
STM32H573	2 Mbytes	Dual	256	64	320	N/A	N/A	4	With hardware crypto: AES, PKA, SAES, and OTFDEC
STM32H563	Up to 2 Mbytes	Dual	256	64	320	N/A	N/A	4	N/A
STM32H562	Up to 2 Mbytes	Dual	256	64	320	N/A	N/A	4	N/A
STM32H553	1 Mbyte	Dual	128	80	96	N/A	N/A	2	With hardware crypto: AES, PKA, SAES, and OTFDEC
STM32H543	Up to 1 Mbyte	Dual	128	80	96	N/A	N/A	2	N/A
STM32H523	Up to 512 Kbytes	Dual	128	80	64	N/A	N/A	2	N/A
STM32H533	512 Kbytes	Dual	128	80	64	N/A	N/A	2	With hardware crypto: AES, PKA, SAES, and OTFDEC
STM32H503	128 Kbytes	Dual	16	16	N/A	N/A	N/A	2	N/A

2.2 System architecture

The STM32H5 devices embed:

- High-speed memories (up to 4 Mbytes of dual-bank flash memory and up to 1536 Kbytes of SRAM).
- A flexible external memory controller (FMC) for devices with packages of 100 pins and more.
- Up to two Octo-SPI memory interfaces and an extensive range of enhanced I/Os and peripherals connected to three APB buses.
- Three AHB buses and a 32-bit multi-AHB bus matrix.

The following table illustrates the bus matrix differences between STM32H5 devices.

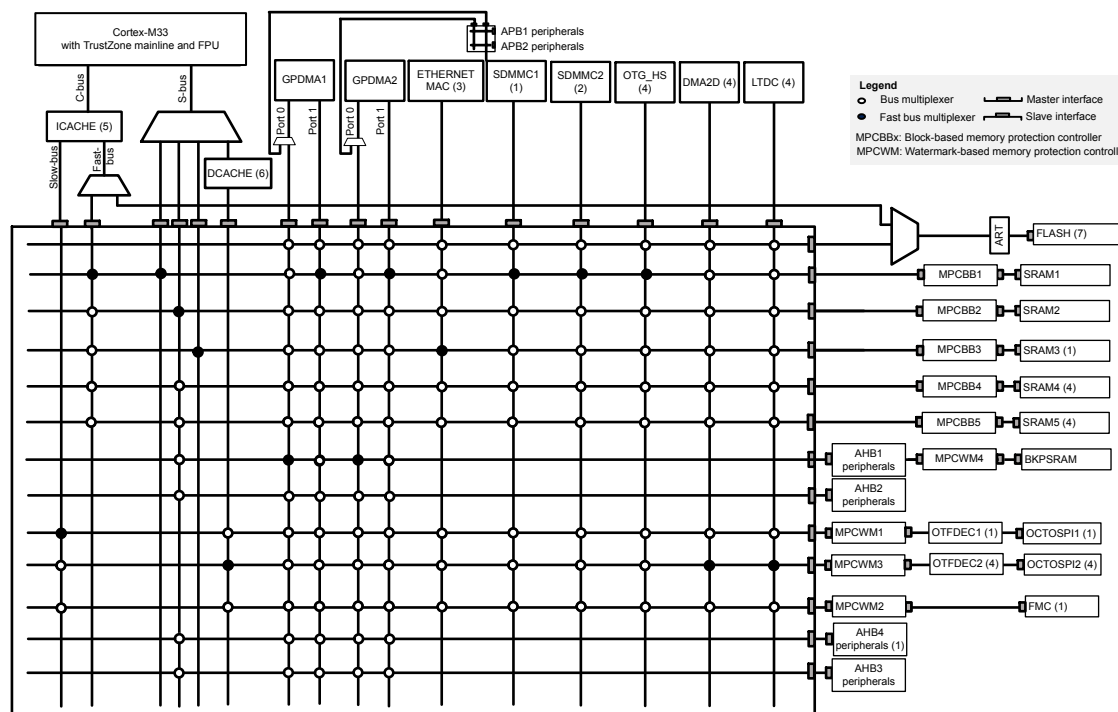
Table 3. Bus matrix

Bus type	STM32H503	STM32H523/533	STM32H543/553	STM32H562 and STM32H563/573	STM32H5Ex/ STM32H5Fx
AHB bus matrix controllers	Up to 7 controllers: Fast C-bus, CPU S-bus for internal memories, GPDMA1 (featuring two controller ports), GPDMA2 (featuring two controller ports).	Up to 11 controllers: Fast C-bus, slow C-bus, CPU S-bus for internal memories, CPU S-bus for external memories, GPDMA1 (featuring two controller ports), GPDMA2 (featuring two controller ports), SDMMC1.	Up to 12 controllers: Fast C-bus, slow C-bus, CPU S-bus for internal memories, CPU S-bus for external memories, GPDMA1 (featuring two controller ports), GPDMA2 (featuring two controller ports), SDMMC1, Ethernet MAC.	Up to 13 controllers: Fast C-bus, slow C-bus, CPU S-bus for internal memories, CPU S-bus for external memories, GPDMA1 (featuring two controller ports), GPDMA2 (featuring two controller ports), SDMMC1, SDMMC2, Ethernet MAC.	Up to 16 controllers: FastC-bus, slow C-bus, CPU S-bus for internal memories, CPU S-bus for external memories, GPDMA1 (featuring two controller ports), GPDMA2 (featuring two controller ports), SDMMC1, SDMMC2, Ethernet MAC, USB OTG HS, DMA2D and LTDC.
AHB bus matrix receivers	Up to 6 receivers: Internal flash memory, SRAM1, SRAM2, AHB1 peripherals (including APB1 and APB2), backup RAM, AHB2 peripherals, AHB3 peripherals.	Up to 10 receivers: Internal flash memory, SRAM1, SRAM2, SRAM3, AHB1 peripherals (including APB1 and APB2), backup RAM, AHB2 peripherals, FMC, OCTOSPI, AHB3 peripherals, AHB4 peripherals.	Up to 10 receivers: Internal flash memory, SRAM1, SRAM2, SRAM3, AHB1 peripherals (including APB1 and APB2), backup RAM, AHB2 peripherals, FMC, OCTOSPI, AHB3 peripherals, AHB4 peripherals.	Up to 10 receivers: Internal flash memory, SRAM1, SRAM2, SRAM3, AHB1 peripherals (including APB1 and APB2), backup RAM, AHB2 peripherals, FMC, OCTOSPI, AHB3 peripherals, AHB4 peripherals.	Up to 13 receivers: Internal flash memory, SRAM1, SRAM2, SRAM3, SRAM4, SRAM5, AHB1 peripherals (including APB1 and APB2), backup RAM, AHB2 peripherals, FMC, OCTOSPI1, OCTOSPI2, AHB3 peripherals, AHB4 peripherals.

The bus matrix provides access from a controller to a receiver, enabling concurrent access and efficient operation even when several high-speed peripherals work simultaneously.

The system architecture of the STM32H5 devices.

Figure 1. STM32H5 devices system architecture



1. Not available on STM32H503 devices
2. Not available on STM32H503, STM32H523/533, and STM32H543/553 devices
3. Available only on STM32H543/553, STM32H563/573, and STM32H5Ex/STM32H5Fx devices
4. Available only on STM32H5E4/F4 and STM32H5E5/F5 devices
5. Icache capacity:
 - 8 Kbytes: STM32H503, STM32H523/533, STM32H543/553, STM32H562, and STM32H563/573 devices
 - 16 Kbytes: STM32H5Ex/STM32H5Fx devices
6. Dcache capacity:
 - 4 Kbytes: STM32H503, STM32H523/533, STM32H543/553, STM32H562, and STM32H563/573 devices
 - 8 Kbytes: STM32H5Ex/STM32H5Fx devices
7. Flash memory capacity:
 - 128 Kbytes: STM32H503 devices
 - Up to 512 Kbytes: STM32H523/533 devices
 - Up to 1 Mbyte: STM32H543/553 devices
 - Up to 2 Mbytes: STM32H562 and STM32H563/573 devices
 - Up to 4 Mbytes: STM32H5Ex/STM32H5Fx devices

DT71848V3

3 Hardware migration

This section presents the package and pinout compatibility details for the hardware migration.

3.1 Package availability

The table below lists the available packages on the STM32H5 devices.

Table 4. Available packages

X = available, N/A = not available

Package (Size in mm x mm)	STM32H503 (LDO)	STM32H523/533 (LDO)	STM32H543/553 (LDO)	STM32H562 and STM32H563/573 (LDO)	STM32H563/573 (SMPS)	STM32H5Ex/ STM32H5Fx (LDO)		STM32H5Ex/ STM32H5Fx (SMPS)	
						LDO	LDO USBPHY	SMPS	SMPS USBPHY
LQFP48 (7 x 7 mm)	X	X	X	N/A	N/A	N/A	N/A	N/A	N/A
LQFP176 (24 x 24 mm)	N/A	N/A	N/A	X	X	X	N/A	X	X
LQFP144 (20 x 20 mm)	N/A	X	X	X	X	X	X	X	N/A
LQFP144-EP (20 x 20 mm)	N/A	N/A	X	N/A	N/A	N/A	N/A	N/A	N/A
LQFP100 (14 x 14 mm)	N/A	X	X	X	X	X	N/A	X	N/A
LQFP100-EP (14 x 14 mm)	N/A	N/A	X	N/A	N/A	N/A	N/A	N/A	N/A
LQFP64 (10 x 10 mm)	X	X	X	X	N/A	N/A	N/A	N/A	N/A
UFQFPN32 (5 x 5 mm)	X	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
UFQFPN48 (7 x 7 mm)	X	X	X	N/A	N/A	N/A	N/A	N/A	N/A
TFBGA255 (13 x 13 mm)	N/A	N/A	N/A	X	X	N/A	N/A	N/A	X
UFBGA176 (10 x 10 mm)	N/A	N/A	N/A	X	X	X	N/A	X	X
UFBGA169 (7 x 7 mm)	N/A	N/A	N/A	X	X	X	N/A	X	N/A
VFQFPN68 (8 x 8 mm)	N/A	N/A	N/A	X	N/A	N/A	N/A	N/A	N/A
UFBGA144 (10 x 10 mm)	N/A	X	X	N/A	N/A	X	N/A	X	N/A
UFBGA100 (7 x 7 mm)	N/A	X	X	N/A	N/A	N/A	N/A	N/A	N/A
WLCSP	WLCSP25 (2.33 x 2.24 mm)	WLCSP39 (2.76 x 2.78 mm)	WLCSP63 (3.01 x 2.97 mm)	N/A	WLCSP80 (3.50 x 3.27 mm)	N/A	N/A	N/A	WLCSP105 (4.4 x 4.4 mm)

3.2 Pinout compatibility

The table below lists the pin- and ballout compatibility between the different STM32H5 series devices.

Table 5. Pin- and ballout compatibility across the STM32H5 series

Package	Package option					Pinout compatibility
	STM32H5Ex/ STM32H5Fx	STM32H562 and STM32H563/573	STM32H543/ 553	STM32H523/ 533	STM32H503	
UFQFPN32	-	-	-	-	LDO	-
UFQFPN48 ⁽¹⁾	-	-	LDO	LDO	LDO	Partial compatibility (refer to Section 3.2.1)
VFQFPN	-	LDO	-	-	-	-
LQFP48	-	-	LDO	LDO	LDO	Full compatibility
LQFP64	-	LDO	-	LDO	LDO	Full compatibility
LQFP100	LDO	LDO	-	LDO	-	Full compatibility
	SMPS	SMPS	-	-	-	Full compatibility
LQFP100-EP	-	-	LDO	-	-	-
LQFP144	LDO	LDO	LDO	LDO	-	Full compatibility
	LDO USBPHY	LDO	LDO	LDO	-	Partial compatibility (refer to Figure 3) ⁽²⁾
	SMPS	SMPS	-	-	-	Full compatibility
LQFP144-EP	-	-	LDO	-	-	-
LQFP176	LDO	LDO	-	-	-	Full compatibility
	SMPS	SMPS	-	-	-	Full compatibility
	SMPS USBPHY	SMPS	-	-	-	Partial compatibility (refer to Figure 4)
UFBGA100	-	-	LDO	LDO	-	Full compatibility
UFBGA144 ⁽¹⁾	LDO	-	LDO	LDO	-	Partial compatibility (refer to Figure 5)
	SMPS	-	-	-	-	-
UFBGA169	LDO	LDO	-	-	-	Full compatibility
	SMPS	SMPS	-	-	-	Full compatibility
UFBGA176	LDO	LDO	-	-	-	Full compatibility
	SMPS	SMPS	-	-	-	Full compatibility
	SMPS USBPHY	SMPS	-	-	-	Partial compatibility (refer to Figure 6)
TFBGA225	SMPS USBPHY	SMPS	-	-	-	No compatibility (refer to Figure 7)
WLCSP	WLCSP105 SMPS USBPHY	WLCSP80 SMPS	WLCSP63 LDO	WLCSP25 LDO	WLCSP39 LDO	-

1. Full compatibility between STM32H543/553 and STM32H523/533

2. Full compatibility between STM32H523/533, STM32H543/553 and STM32H562, STM32H563/573

For more details, refer to the corresponding application note [4].

The following sections show the package pin- and ballout figures and difference tables.

3.2.1 UFQFPN48 LDO package

Figure 2. UFQFPN48 pinout

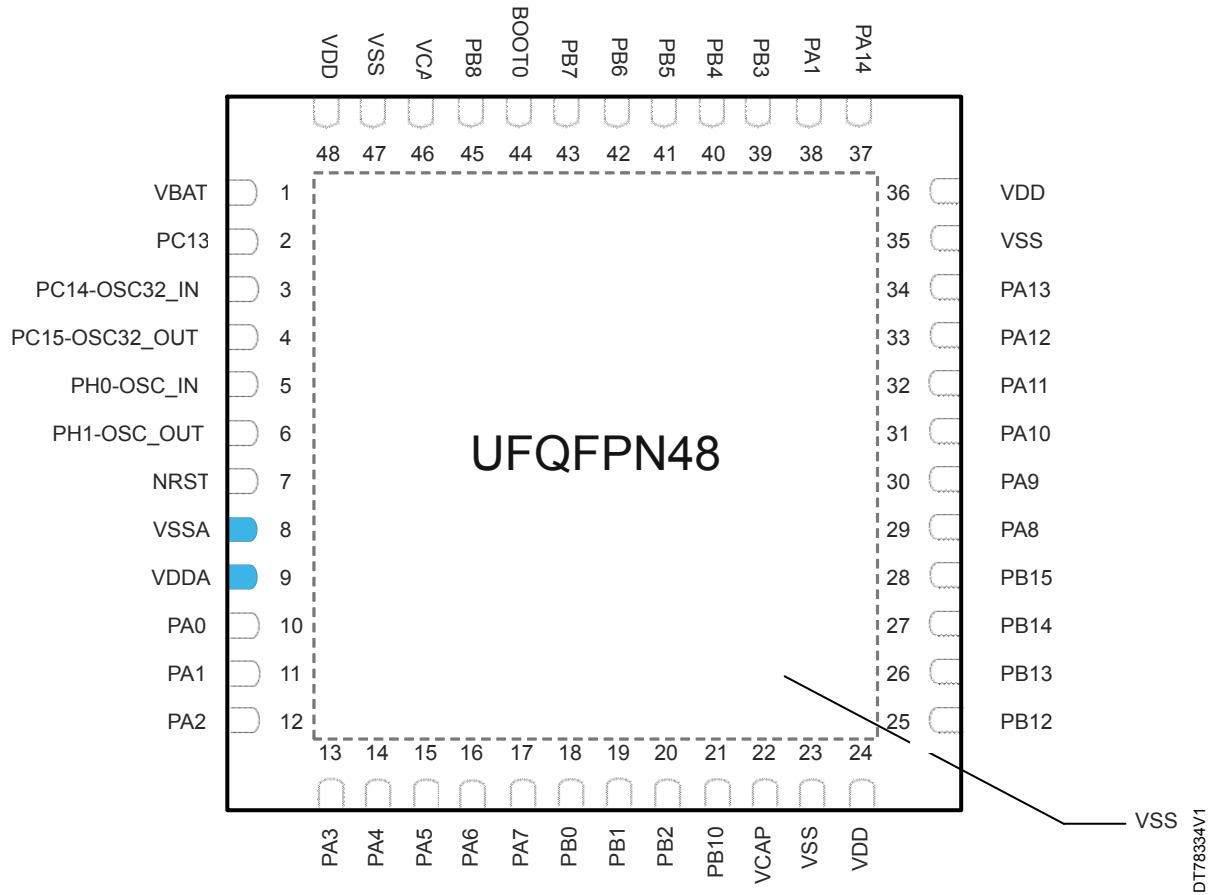
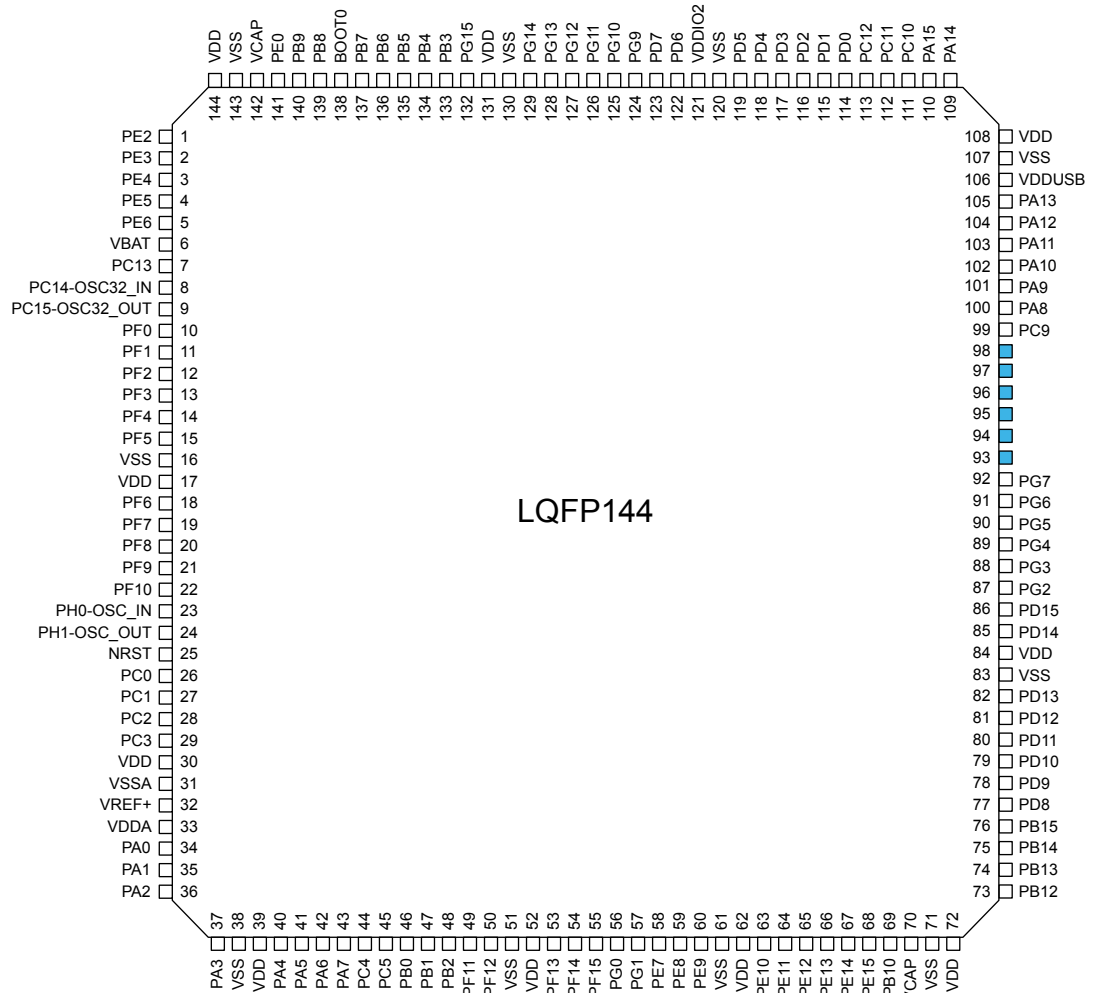


Table 6. LQFP48 pinout differences

STM32H523/533, and STM32H543/553LDO	STM32H5Ex/STM32H5Fx LDO USBPHY
8 9 VSSA VDDA	8 9 VSSA/VREF- VDDA/VREF+

3.2.2 LQFP144 LDO package

Figure 3. LQFP144 pinout



DT74885V1

Table 7. LQFP144 pinout differences

STM32H563/573, STM32H543/553 and STM32H523/533, STM32H562, and STM32H5Ex/STM32H5Fx LDO	STM32H5Ex/STM32H5Fx LDO USBPHY
98 PC8 97 PC7 96 PC6 95 VDD 94 VSS 93 PG8	98 VSSUSB 97 OTG_HS_DP 96 OTG_HS_DM 95 VDD11USB 94 VDD 93 VSS

3.2.3 LQFP176 SMPS package

Figure 4. LQFP176 pinout

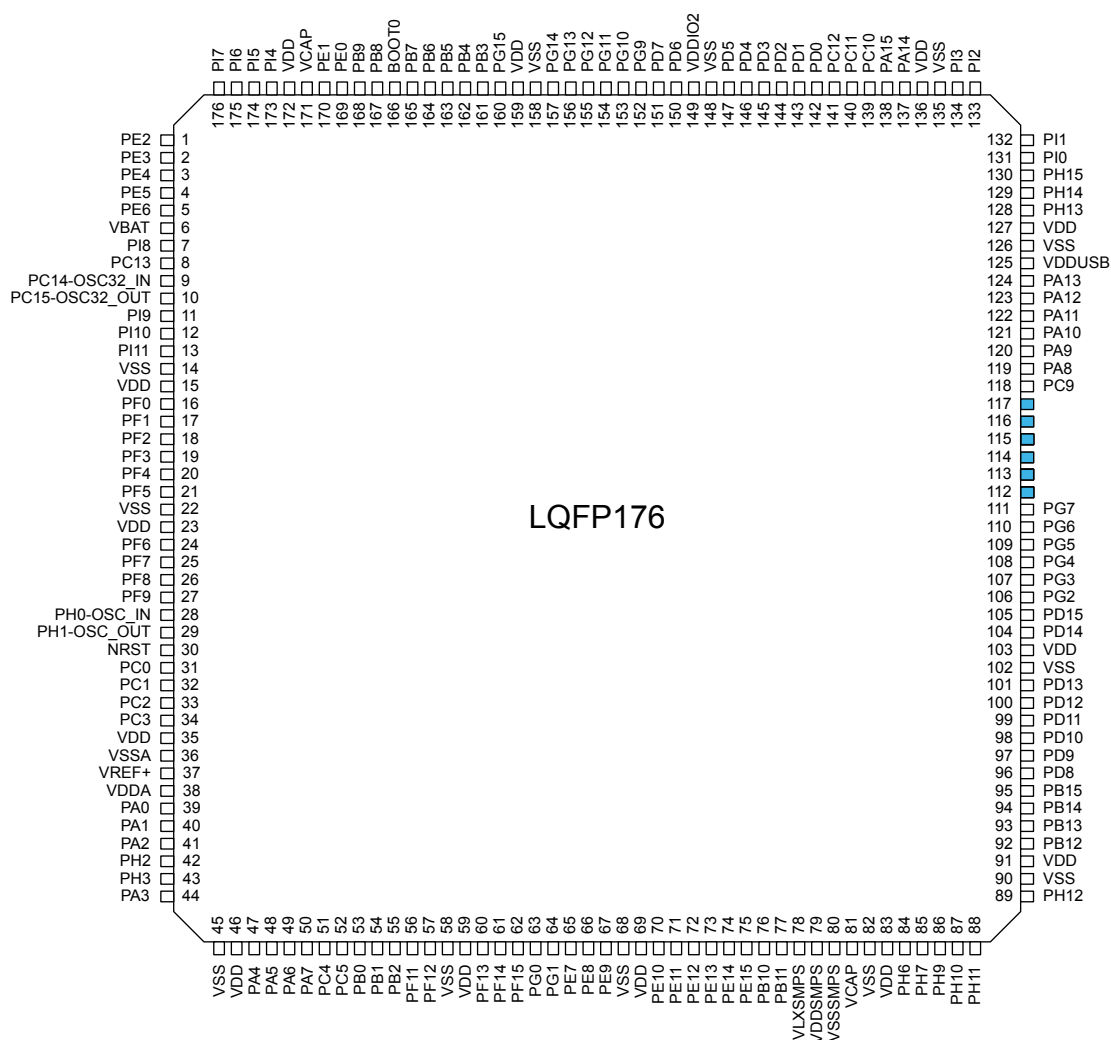
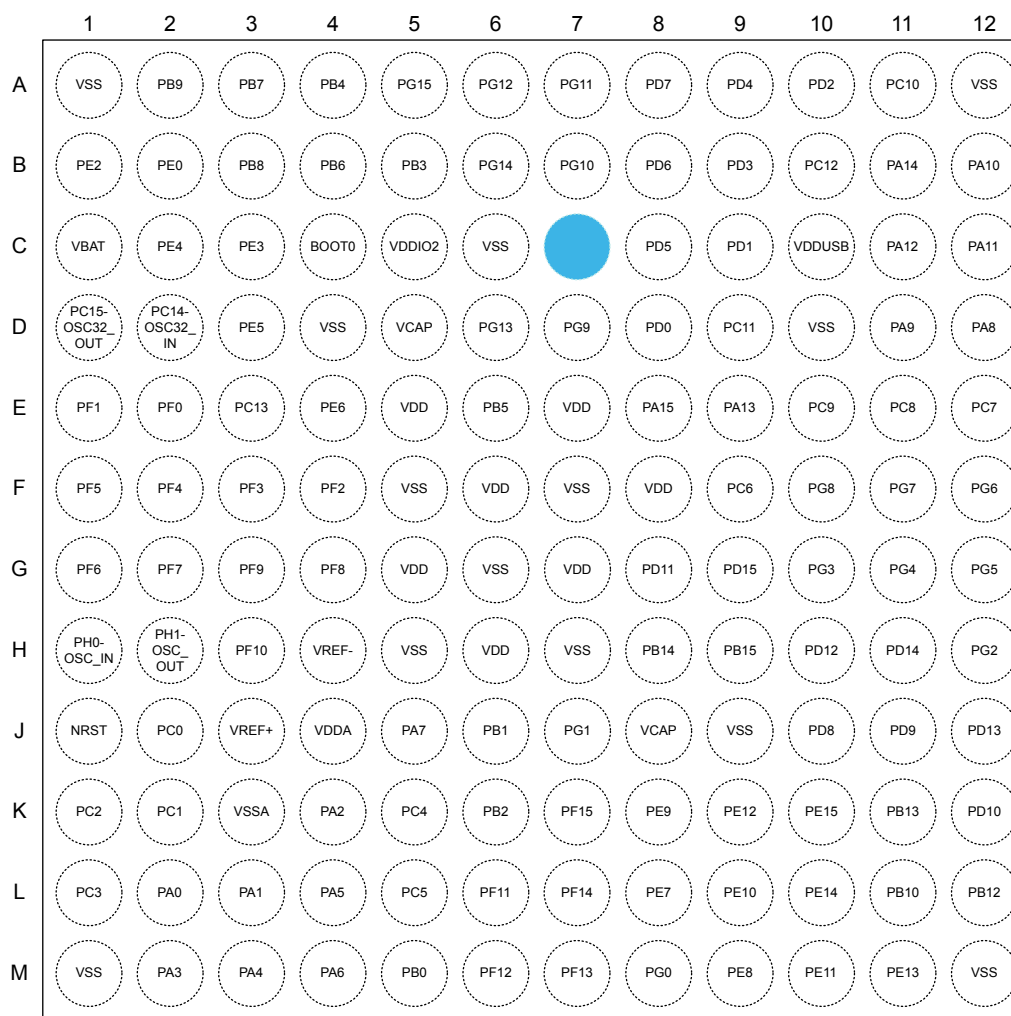


Table 8. LQFP176 pinout differences

STM32H563/573, STM32H562, and STM32H5Ex/STM32H5Fx SMPS	STM32H5Ex/STM32H5Fx SMPS USBPHY
117 PC8 116 PC7 115 PC6 114 VDD 113 VSS 112 PG8	117 VSSUSB 116 OTG_HS_DP 115 OTG_HS_DM 114 VDD11USB 113 VDD 112 VSS

3.2.4 UFBGA144 LDO package

Figure 5. UFBGA144 ballout



DT74887V1

Table 9. UFBGA144 ballout differences

STM32H523/533 and STM32H543/553 LDO	STM32H5Ex/STM32H5Fx LDO

3.2.5 UFBGA176 SMPS package

Figure 6. UFBGA176 ballout

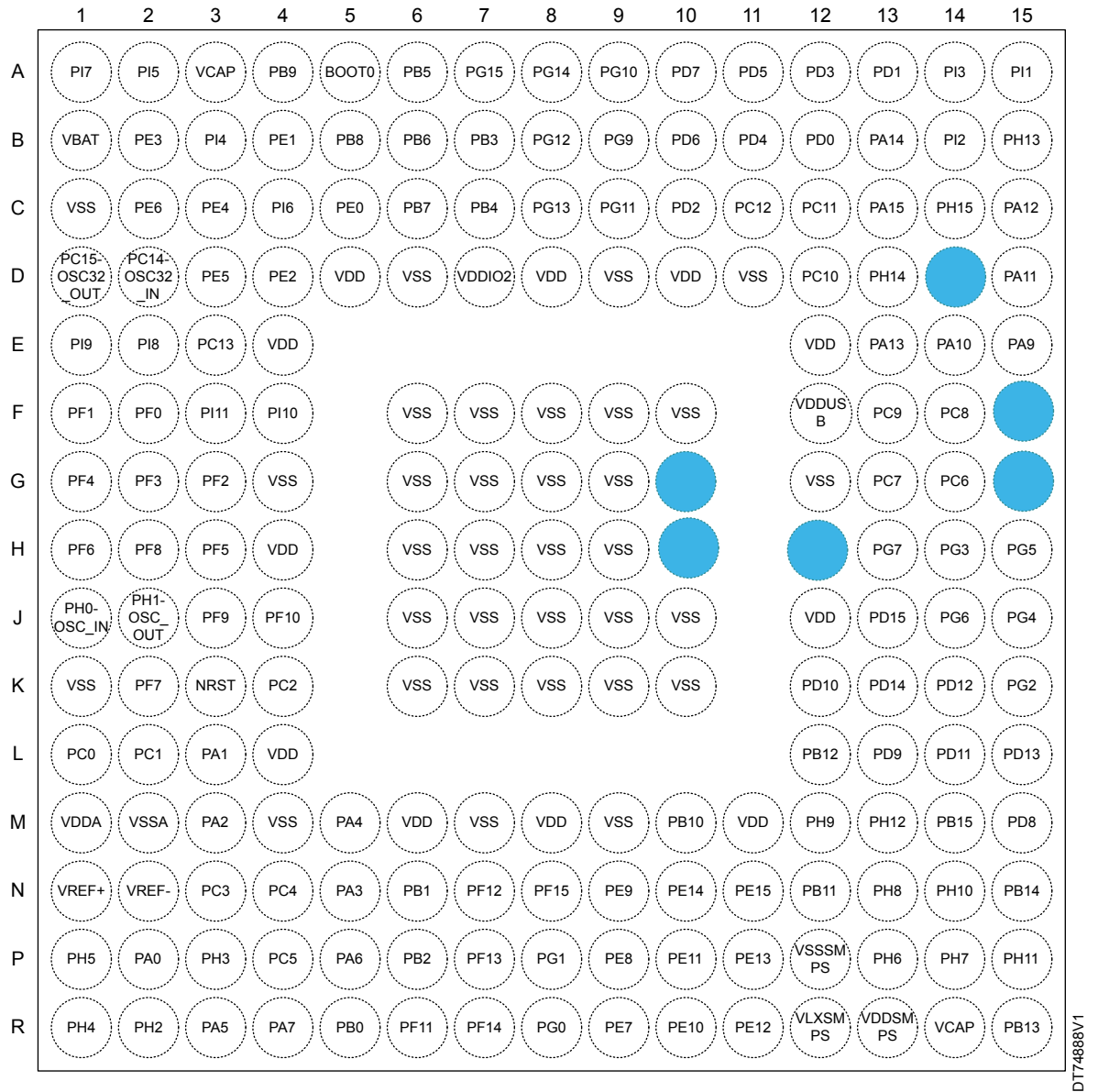
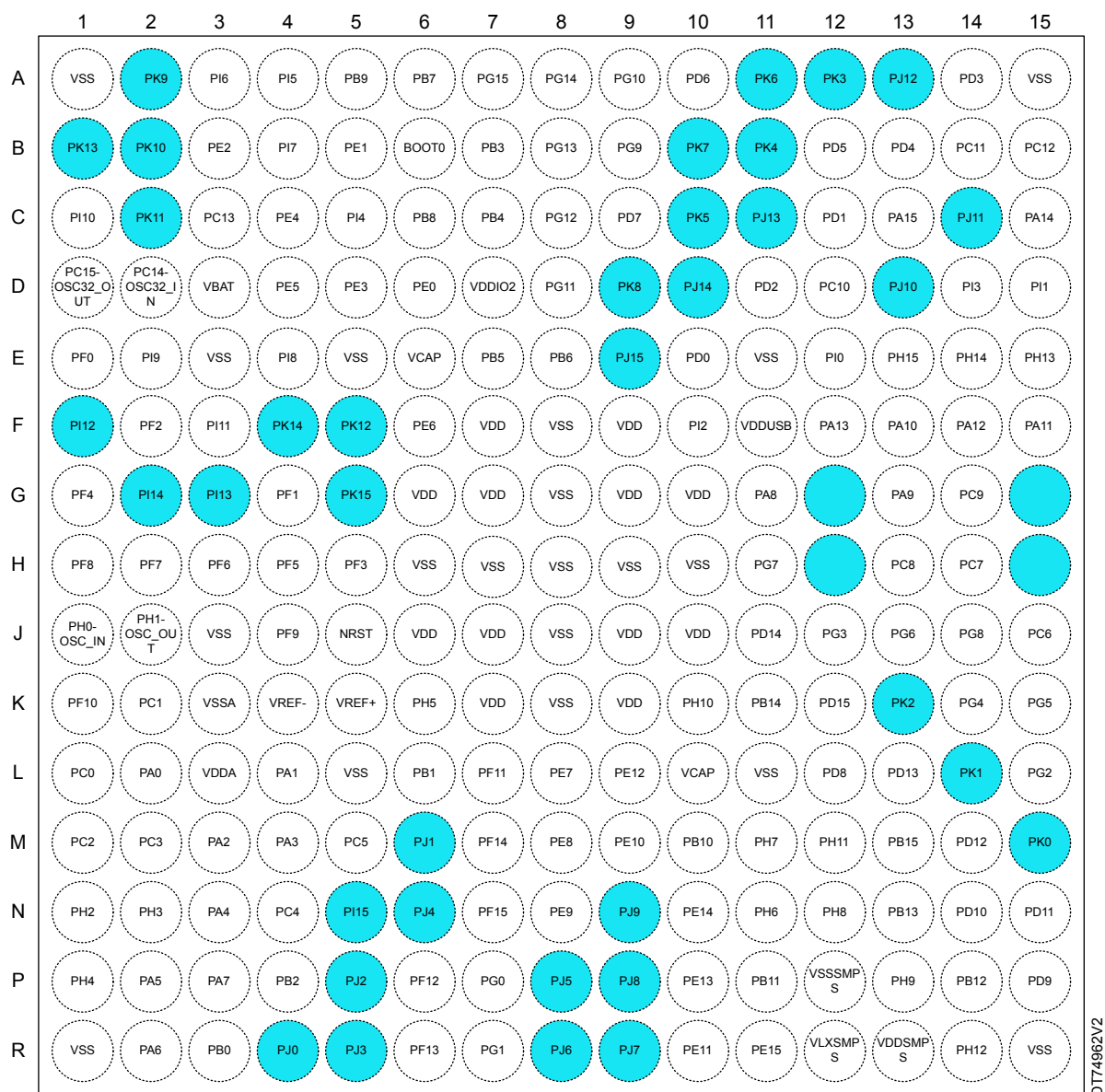


Table 10. UFBGA176 ballout differences

STM32H563/573, STM32H562, and STM32H5Ex/STM32H5Fx SMPS	STM32H5Ex/STM32H5Fx SMPS USBPHY

3.2.6 TFBGA225 SMPS USBPHY package

The STM32H5Ex/STM32H5Fx have additional GPIOs, located on Ports I, J, and K, which are not present on the STM32H56x/STM32H57x. Specifically, Port I located on PI[15:12], as well as all pins on Ports PJ[15:0] and PK[15:0], are exclusive to STM32H5Ex/STM32H5Fx. The TFBGA225 package version of the STM32H56x/STM32H57x devices lack these GPIOs but include VSS pins instead.

Figure 7. TFBGA225 SMPS USBPHY ballout


DT74962V2

Table 11. TFBGA225 ballout differences

STM32H563/573, and STM32H562 SMPS				STM32H5Ex/STM32H5Fx SMPS USBPHY			
PA13	PA10	PA12	PA11	PA13	PA10	PA12	PA11
VSS	PA9	PC9	VSS	VSSUSB	PA9	PC9	OTG_HS_DP
VSS	PC8	PC7	VSS	VSS11 USB	PC8	PC7	OTG_HS_DM
PG3	PG6	PG8	PC6	PG3	PG6	PG8	PC6

4 Boot mode compatibility

4.1 System bootloader

The system bootloader is in the system memory, programmed by STMicroelectronics during the production. It is used to reprogram the flash memory using one of the following serial interfaces.

The following table shows the communication peripherals supported by the system bootloader. For more details, refer to the application note *STM32 microcontroller system memory boot mode* (AN2606).

Table 12. Bootloader communication peripherals

System bootloader peripherals	STM32H503 I/O pin	STM32H523/533 I/O pin	STM32H543/553 I/O pin	STM32H562 and STM32H563/573 I/O pin	STM32H5Ex/ STM32H5Fx
DFU	PA11/PA12				PA11/PA12 (OTG_FS)
	N/A				OTG_HS_DP / OTG_HS_DM
USART1	PA10/PA9				
USART2	PA15/PA5	PA3/PA2	PA3/PA2 ⁽¹⁾	PA3/PA2	PA2/PA3
USART3	PA3/PA4	PD9/PD8 ⁽¹⁾	PD9/PD8 ⁽¹⁾	PD9/PD8	PD8/PD9
CAN	FDCAN1 (PB5/PB15)	FDCANx (PB5/PB13) ⁽²⁾	FDCAN2 (PB5/PB13)	FDCAN2 (PB5/PB13)	N/A
I2C1	N/A	PB8/PB9 ⁽³⁾	PB8/PB9 ⁽³⁾	PB8/PB9 ⁽³⁾	N/A
I2C2	PB3/PB4	N/A	N/A	N/A	N/A
I2C3	N/A	PA8/PC9 ⁽⁴⁾	PA8/PB4	PA8/PC9	PA8/PC9
I2C4	N/A	N/A	N/A	PD12/PD13	PD12/PD13
I3C1	PB6/PB7	PB6 ⁽⁵⁾ /PB7	PB6/PB7	PB6/PB7	PB6/PB7
SPI1	PA7/PA0/PA8/PB8	PA7/PA6/PA5/PA4	PA7/PA6/PA5/PA4	PA7/PA6/PA5/PA4	N/A
SPI2	PB1/PB14/PB10/PB12	PC1 ⁽⁶⁾ /PB14/PB10/PB12 or PB15/PB14/PB10/PB12	PC1/PB14/PB10/PB12 or PB15/PB14/PB10/PB12	PC1/PB14/PB10/PB12	N/A
SPI3	PC12/PC11/PC10/PD2	PC12/PC11/PC10/PA15 ⁽³⁾	PC12/PC11/PC10/PA15	PC12/PC11/PC10/PA15 ⁽⁷⁾	N/A

1. Only for LQFP100, LQPF100-EP, LQFP144, LQFP144-EP, UFBGA144, and UFBGA100.

2. On STM32H5xx devices, FDCAN bootloader does not use an external quartz, and uses HSI and PLL instead.

3. Not available for LQFP48, LQFP64, UFQFN48, WLCSP63, and WLCSP39.

4. PC9 is replaced by PB4 on LQFP48, UFQFN48, and WLCSP39.

5. PB6 is replaced by PB8 on LQFP48, UFQFN48, and WLCSP39.

6. PC1 is replaced by PB15 on LQFP48, UFQFN48, and WLCSP39.

7. Not available for LQFP48, UFQFN48

5 Peripheral migration

5.1 Cross-compatibility between STM32 products

STM32 microcontrollers embed a set of peripherals that can be classified in the following groups:

- Group 1: peripherals by definition common to all products
These peripherals are identical, so they have the same structure, registers, and control bits. There is no need to perform any firmware changes to keep the same functionality at the application level after migration. All features and behavior remain the same.
- Group 2: peripherals shared by all products but with only minor differences (in general to support new features)
The migration from one product to another is very straightforward and does not require any significant new development efforts.
- Group 3: peripherals that have considerable changes from one product to another (new architecture or new features, for example)
For this group of peripherals, the migration requires new developments at the application level.

The following table summarizes the available peripherals in STM32H5 series microcontrollers as well as their compatibility.

Table 13. STM32 peripheral compatibility between products

Peripherals		STM32H503	STM32H523/533	STM32H543/553	STM32H562 and STM32H563/573	STM32H5Ex/ STM32H5Fx
Core		Arm® Cortex®-M33				
Maximum CPU frequency		Up to 250 MHz				
Flash memory		128 Kbytes	512 Kbytes	1 Mbytes	2 Mbytes	4 Mbytes
SRAMs	System	32 Kbytes (16 + 16)	272 Kbytes (128 + 80 + 64)	304 Kbytes (128+80+96)	640 Kbytes (256 + 64 + 320)	1536 Kbytes (256 + 128 + 384 + 384 + 384)
	Backup	2 Kbytes	2 Kbytes	2 Kbytes	4 Kbytes	4 Kbytes
Timers	General purpose	1 (32 bits) and 1 (16 bits)	2 (32 bits) and 4 (16 bits)	2 (32 bits) and 4 (16 bits)	2 (32 bits) and 8 (16 bits)	2 (32 bits) and 8 (16 bits)
	Advanced control	1 (16 bits)	2 (16 bits)	2 (16 bits)	2 (16 bits)	2 (16 bits)
	Basic	2 (16 bits)	2 (16 bits)	2 (16 bits)	2 (16 bits)	2 (16 bits)
	Low power	2 (16 bits)	2 (16 bits)	2 (16 bits)	6 (16 bits)	6 (16 bits)
	SysTick timer	1	2	2	2	2
	Watchdog timers (independent, window)	2	2	2	2	2
Communication interfaces	SPI/I2S	Up to 3 SPIs, including three multiplexed with full-duplex I2S, and up to 3 additional SPIs from 3 USARTs when configured in	Up to 4 SPIs, including three multiplexed with full-duplex I2S	Up to 4 SPIs, including three multiplexed in full-duplex I2S audio class accuracy via internal audio PLL or external clock, and up to four additional SPIs from four USARTs when configured in Synchronous	Up to 6 SPIs, including three multiplexed with full-duplex I2S and up to 5x additional SPIs from 5 USARTs when configured in synchronous mode (one additional SPI with OCTOSPI)	Up to 6 SPIs, including three multiplexed with full-duplex I2S and up to 5 additional SPIs from 5 USARTs when configured in synchronous mode (one

Peripherals		STM32H503	STM32H523/533	STM32H543/553	STM32H562 and STM32H563/573	STM32H5Ex/ STM32H5Fx
Communication interfaces		synchronous mode		mode (one additional SPI with OctoSPI)		additional SPI with OctoSPI).
	I2C	2 (Fm+ interfaces SMBus/ PMBus®)	Up to 3 (Sm and Fm interfaces SMBus/PMBus®)	Up to 3 I ² Cs Fm+ (SMBus/PMBus®)	4 (Sm, Fm, and Fm+ interfaces (SMBus/PMBus®)	4 (Sm, Fm, and Fm+ interfaces (SMBus/ PMBus®)
	I3C	2	2	2	1	2
	USART/UART	Up to 3 USARTs	4 / 2	4 / 4	6 / 6	6 / 6
	LPUART	1	1	1	1	1
	USB_FS	Yes	Yes	Yes	Yes	No
	OTG_FS	No	No	No	No	Yes
	OTG_HS	No	No	No	No	Yes ⁽⁴⁾
	UCPD	No	Yes	Yes	Yes	Yes
	CAN	1 FDCAN	x2 FDCAN	x2 FDCAN	x2 FDCAN	x3 FDCAN
	SAI	No	No	No	2	2
	SDIO/SDMMC	No	1	1	2	2
	DCMI	No	Yes	No	Yes	Yes
	PSSI	No	Yes	No	Yes	Yes
	Ethernet	No	No	Yes (1x Ethernet 10/100)	Yes ⁽¹⁾ (1x Ethernet 10/100)	Yes (1x Ethernet 10/100)
Flexible memory controller (FMC) ⁽⁶⁾		No	Yes (8, 16-bit data bus width, no SDRAM)	Yes (8, 16-bit data bus width with SDRAM)	Yes (8, 16-bit data bus width with SDRAM)	Yes (8, 16, 32-bit data bus width with SDRAM) ⁽⁵⁾
OCTOSPI		No	1	1	1	2
DMA		2 GPDMA (privileged / unprivileged support / linked-list)	2 GPDMA (featuring two controller ports), TrustZone® support / linked-list	2 GPDMA (featuring two controller ports), TrustZone® support / linked-list	2 GPDMA (featuring two controller ports), TrustZone® support / linked-list	2 GPDMA (featuring two controller ports), TrustZone® support / linked-list
CORDIC coprocessor		No	No	Yes	Yes	Yes
Filter mathematical accelerator (FMAC)		No	No	No	Yes	Yes
Real-time clock (RTC)		Yes	Yes	Yes	Yes	Yes
Random number generator (RNG)		Yes	Yes	Yes	Yes	Yes
SAES ⁽²⁾ , AES		No	Yes	Yes	Yes	Yes
Coupling and chaining bridge (CCB)		No	No	Yes	No	Yes
Public key accelerator (PKA) ⁽²⁾		No	Yes	Yes	Yes	Yes
HASH (SHA-512)		SHA-1, SHA-2 (SHA-256)	SHA-1, SHA-2	SHA-1, SHA-2, SHA-3	SHA-1, SHA-2	SHA-1, SHA-2, SHA-3

Peripherals		STM32H503	STM32H523/533	STM32H543/553	STM32H562 and STM32H563/573	STM32H5Ex/ STM32H5Fx
On-the-fly decryption engine (OTFDEC)		No	Yes	Yes	Yes ⁽³⁾	Yes
GPIOs	Available GPIOs	Up to 49	Up to 112	Up to 112	Up to 140	Up to 176
	GPIOs supplied by VDDIO2	Up to 9	Up to 10	Up to 10	Up to 10	Up to 10
	Wakeup pins	Up to 5	Up to 7	Up to 7	Up to 10	Up to 10
ADC (12 bits)		1 (12-bit ADC, up to 2.5 MSPS in 12-bit) Up to 16 channels	2 (12-bit ADC, up to 5 MSPS) Up to 20 channels	3 (12-bit ADC, up to 5 MSPS) Up to 30 channels	2 (12-bit ADC, up to 5 MSPS) Up to 20 channels	3 (12-bit ADC, up to 5 MSPS) Up to 30 channels
DAC (12 bits)		1x DAC (2 channels)				
Comparator (COMP)		1	No	2	No	2
Operation amplifier (OPAMP)		1	No	No	No	1
CRC		Yes				
Multifunction Digital Filter (MDF)		No	No	No	No	Yes ⁽⁷⁾
Audio Digital Filter (ADF)		No	No	No	No	1x ADF
Graphic accelerators	JPEG codec	No	No	No	No	Yes
	LTDC (LCD-TFT display controller)	No	No	No	No	Yes
	Chrom-ART2 Accelerator DMA2D	No	No	No	No	Yes
	GFXTIM	No	No	No	No	Yes
Programmable logic array (PLAY)		No	No	Yes	No	Yes
HDMI CEC		No	Yes	Yes	Yes	Yes
Tamper pins	Passive	Up to 2	Up to 8	Up to 8	Up to 8	8
	Active (shared output)	Up to 2	Up to 7	Up to 7	Up to 7	8
	Active (independent out)	Up to 2	Up to 7	Up to 7	Up to 7	Up to 5
Operating temperatures		Ambient operating temperature: -40°C to +85°C / 105°C, up to +125°C at low dissipation Junction temperature: -40°C to +130°C	Ambient operating temperature: -40°C to +85°C, up to +125°C Junction temperature: -40°C to +130°C	Ambient operating temperature: -40°C to 85°C, -40°C to 105°C, -40°C to 125°C Junction temperature: -40°C to 140°C	Ambient operating temperature: -40°C to +85°C / -40°C to +125°C Junction temperature: -40°C to +130°C	Ambient operating temperature: -40°C to +85°C / -40°C to +105°C Junction temperature: -40 to 130 °C
Operating voltage		1.71 to 3.6 V				

Peripherals	STM32H503	STM32H523/533	STM32H543/553	STM32H562 and STM32H563/573	STM32H5Ex/STM32H5Fx
Internal voltage reference buffer	No	Yes	Yes	Yes	Yes

1. Not available on STM32H562 devices.
2. Available only on STM32H533, STM32H553, STM32H573, and STM32H5F devices.
3. OTFDEC is not supported on STM32H56x devices.
4. Only for LQFP144, WLCSP105, LQFP176, UFBGA176, and TFBGA255 packages.
5. NOR/RAM SDRAM not available for LQFP100 and WLCSP105 packages.
6. Available only in packages with 100 pins and more.
7. with 6 filters (DFLT) and serial interfaces (SITF)

Note: For further details, refer to the product datasheets.

Revision history

Table 14. Document revision history

Date	Version	Changes
20-Mar-2024	1	Initial release.
24-Feb-2026	2	Added: - Section 3.2.6: TFBGA225 SMPS USBPHY package Updated: - Update the product reference from STM32H563/STM32H573 to STM32H563/H573 throughout the document - Introduction - Section 2: STM32H5 MCU overview - Section 2.1: Memory availability - Table 2. Memory availability - Section 2.2: System architecture - Table 3. Bus matrix - Section 3.1: Package availability - Table 4. Available packages - Section 3.2: Pinout compatibility - Table 5. Pin- and ballout compatibility across the STM32H5 series - Table 12. Bootloader communication peripherals - Table 13. STM32 peripheral compatibility between products
25-Mar-2026	3	Status change from ST Restricted to public release Added: - Section 3.2.2: LQFP144 LDO package - Section 3.2.3: LQFP176 SMPS package - Section 3.2.4: UFBGA144 LDO package - Section 3.2.5: UFBGA176 SMPS package Updated: - Table 1. Reference documents - Figure 1. STM32H5 devices system architecture
23-Jun-2026	4	Added: - Support for STM32H523/533 MCU devices in the whole document. - Section 3.2.1: UFQFPN48 LDO package Updated: - Section Introduction - Section 1: General information - Table 1. Reference documents - Section 2: STM32H5 MCU overview - Section 2.1: Memory availability - Table 2. Memory availability - Section 2.2: System architecture - Table 3. Bus matrix - Table 4. Available packages - Table 5. Pin- and ballout compatibility across the STM32H5 series - Table 7. LQFP144 pinout differences - Table 12. Bootloader communication peripherals - Section 5.1: Cross-compatibility between STM32 products

Contents

1	General information	2
2	STM32H5 MCU overview	3
2.1	Memory availability	3
2.2	System architecture	3
3	Hardware migration	6
3.1	Package availability	6
3.2	Pinout compatibility	7
3.2.1	UFQFPN48 LDO package	8
3.2.2	LQFP144 LDO package	9
3.2.3	LQFP176 SMPS package	10
3.2.4	UFBGA144 LDO package	11
3.2.5	UFBGA176 SMPS package	12
3.2.6	TFBGA225 SMPS USBPHY package	13
4	Boot mode compatibility	16
4.1	System bootloader	16
5	Peripheral migration	17
5.1	Cross-compatibility between STM32 products	17
	Revision history	21
	List of tables	23
	List of figures	24

List of tables

Table 1.	Reference documents	2
Table 2.	Memory availability.	3
Table 3.	Bus matrix.	4
Table 4.	Available packages	6
Table 5.	Pin- and ballout compatibility across the STM32H5 series	7
Table 6.	LQFP48 pinout differences	8
Table 7.	LQFP144 pinout differences	9
Table 8.	LQFP176 pinout differences	10
Table 9.	UFBGA144 ballout differences.	11
Table 10.	UFBGA176 ballout differences.	13
Table 11.	TFBGA225 ballout differences	15
Table 12.	Bootloader communication peripherals	16
Table 13.	STM32 peripheral compatibility between products	17
Table 14.	Document revision history	21

List of figures

Figure 1.	STM32H5 devices system architecture	5
Figure 2.	UFQFPN48 pinout	8
Figure 3.	LQFP144 pinout	9
Figure 4.	LQFP176 pinout	10
Figure 5.	UFBGA144 ballout	11
Figure 6.	UFBGA176 ballout	12
Figure 7.	TFBGA225 SMPS USBPHY ballout.	14

IMPORTANT NOTICE – READ CAREFULLY

STMicroelectronics NV and its subsidiaries ("ST") reserve the right to make changes, corrections, enhancements, modifications, and improvements to ST products and/or to this document at any time without notice.

In the event of any conflict between the provisions of this document and the provisions of any contractual arrangement in force between the purchasers and ST, the provisions of such contractual arrangement shall prevail.

The purchasers should obtain the latest relevant information on ST products before placing orders. ST products are sold pursuant to ST's terms and conditions of sale in place at the time of order acknowledgment.

The purchasers are solely responsible for the choice, selection, and use of ST products and ST assumes no liability for application assistance or the design of the purchasers' products.

No license, express or implied, to any intellectual property right is granted by ST herein.

Resale of ST products with provisions different from the information set forth herein shall void any warranty granted by ST for such product.

If the purchasers identify an ST product that meets their functional and performance requirements but that is not designated for the purchasers' market segment, the purchasers shall contact ST for more information.

ST and the ST logo are trademarks of ST. For additional information about ST trademarks, refer to www.st.com/trademarks. All other product or service names are the property of their respective owners.

Information in this document supersedes and replaces information previously supplied in any prior versions of this document.

© 2026 STMicroelectronics – All rights reserved